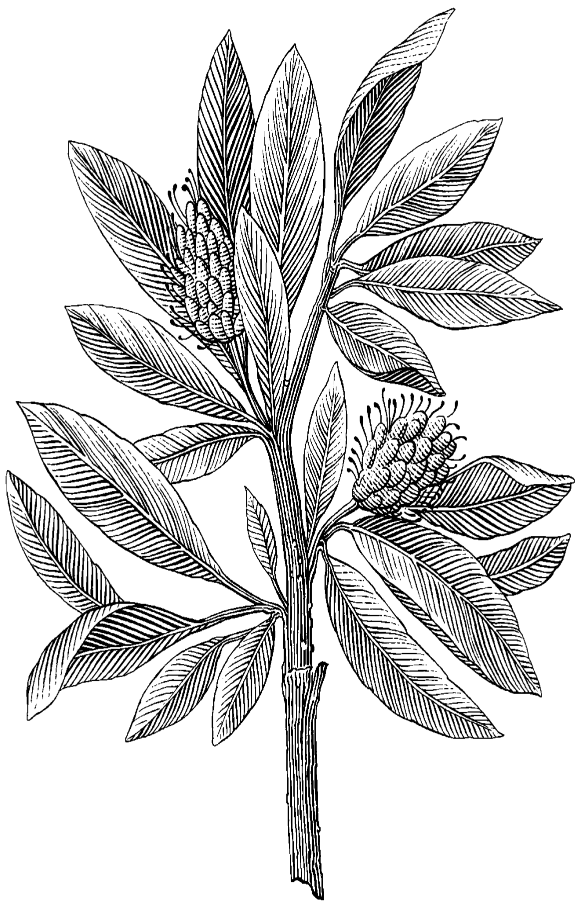
Linnéuniversitetet
Kalmar Växjö

Report

Project
1IK172

Author: Elena Berg - Matteo Pinzani
Semester: Spring 2026
Submitted: 2026-05-20
Email eb22ag@student.lnu.se - mp224qb@student.lnu.se



1. Dataset Selection

1.1 Data Source and Project Objective

For this project, we selected the '**Flight Price Prediction**' dataset, a publicly available open-source dataset uploaded on Kaggle. The raw data represents secondary data extracted from the Indian flight booking platform "Ease My Trip" using the Octoparse web scraping tool.

The primary objective of this study is to analyze the factors that influence domestic flight ticket prices in India and to create predictive regression models. By exploring this dataset, we want to uncover meaningful patterns and insights, such as the impact of booking proximity or seat class that can assist consumers in optimizing their travel expenses and help airlines understand demand dynamics.

1.2 Dataset Characteristics

The selected dataset contains approximately **300,000 distinct flight booking options** captured over a 50-day window. It covers flight travel between India's top 6 metropolitan cities: Delhi, Mumbai, Bangalore, Kolkata, Hyderabad, and Chennai. The dataset includes **11 features** (both categorical and numeric variables):

#	Attribute Name	Type	Description
1	Airline	Categorical (Nominal)	The name of the operating airline company (6 unique values: Vistara, Air India, Indigo, etc)
2	Flight	Categorical (Nominal)	The specified plane flight code (1,561 unique values)
3	Source_City	Categorical (Nominal)	The departure city from which the plane takes off (6 unique cities)
4	Departure_Time	Categorical (Ordinal)	Binned time intervals representing the departure period (6 labels, such as Early Morning, Morning, Night)
5	Stops	Categorical (Ordinal)	The number of stops between source and destination (3 values: zero, one, or two+ stops)
6	Arrival_Time	Categorical (Ordinal)	Binned time intervals representing when the plane lands (6 labels).
7	Destination_City	Categorical (Nominal)	The airport destination where the flight lands (6 unique cities)
8	Class	Categorical (Binary)	The cabin class of the ticket option (Economy or Business)

9	Duration	Continuous Numeric	The total flight travel time between cities measured in hours
10	Days_Left	Discrete Numeric	A derived feature calculating the days remaining between the booking date and the departure date
11	Price (Target)	Continuous Numeric	The actual cost of the ticket in Indian Rupees. This is the dependent variable we want to predict

2. Data Preprocessing

The preprocessing phase is critical to ensure data quality and model reliability, following the CRISP-DM framework. Given the large scale of the original dataset (over 300,000 instances), the following steps were performed.

2.1 Data Sampling and Initial Cleaning

1. *Manual ID Addition*

   An `id` column was manually added to the CSV file to ensure unique tracking of each instance during the initial exploration. This step also resolved native parsing errors encountered during Weka's initial dataset import.

2. *Sampling*

   To optimize computational performance in Weka and avoid memory errors, we applied the `Resample` filter. We extracted a 5% representative sample (approximately 15,000 instances), maintaining the original distribution.

3. *Data Integrity Check*

   We performed a rigorous check for missing values and invalid data (e.g. negative prices or impossible durations). No missing or invalid values were found in this specific dataset.

   - *Note on hypothetical cleaning*

     If missing values had been present, we would have applied the `ReplaceMissingValues` filter (using Mean/Mode imputation) or removed the affected rows if the missingness exceeded a specific limit. For invalid numerical values, we would have applied a custom `RemoveWithValues` filter.

2.2 Dimensionality Reduction and Noise Removal

4. *ID Removal*

   The `id` attribute was removed as it lacks predictive power and would lead to overfitting.

5. *Flight Code Removal*

We identified the flight attribute as redundant noise. With 1,561 unique flight codes, keeping this attribute would have caused extreme data sparsity and the "curse of dimensionality" during the encoding process. Removing it allowed for a more interpretable model focused on high-level features like class.

6. *Outlier Treatment*

   Using the `InterquartileRange` (IQR) filter, we analyzed the dataset for instances that deviated from the normal distribution. The filter flags anomalies by creating two auxiliary attributes: 'Outlier' (values beyond 1.5x IQR) and 'ExtremeValue' (values beyond 3.0x IQR).

   - Detection

     Out of the 15,007 sampled instances, the filter identified exactly 1 'Outlier' and 0 'ExtremeValues'.

   - Removal

     To physically remove this anomaly, we applied the `RemoveWithValues` filter, specifically targeting the 'yes' value of the newly created 'Outlier' attribute.

   - Cleanup

     Finally, the auxiliary 'Outlier' and 'ExtremeValue' columns were deleted. This ensured the model would not use these temporary tags as predictive features, keeping the regression coefficients unbiased.

2.3 Data Transformation

7. *Nominal to Binary (One-Hot Encoding)*

   Categorical variables with no intrinsic order (airline, source_city, destination_city) were transformed into binary indicators using the `NominalToBinary` filter. This prevents the model from assuming a false numerical hierarchy between different cities or airlines.

8. *Logical Reordering and Ordinal Mapping*

   A critical challenge was identified regarding time-based attributes (departure_time and arrival_time). By default, Weka assigns numerical indices to nominal labels based on their appearance in the CSV or alphabetical order. However, these attributes are ordinal in nature and follow a strict chronological sequence.

   - Correction step

     To prevent the model from learning incorrect temporal relationships, we used the `SwapValues` filter. This allowed us to manually reorder the labels into their natural chronological progression (Late_Night, Early_Morning, Morning, Afternoon, Evening, Night).

   - Transformation

     Only after establishing this logical hierarchy we applied the `OrdinalToNumeric` filter (also to stops). This ensured that the resulting integers (1 to 6) represent a meaningful trend, allowing the regression algorithms to correctly identify how the progression of the day affects flight pricing.

9. *Normalization*

We applied the `Normalize` filter to all numeric attributes, scaling them to a [0,1] range. We chose Normalization over Standardization because our preliminary tests showed that preserving the exact distribution boundaries produced lower Mean Absolute Error (MAE) for our specific regression models (Random Forest, SMOreg, and Linear Regression).

2.4 Attribute Selection

10. *Correlation-based Feature Selection*

To identify the most powerful predictors and further reduce dimensionality, we applied the `CfsSubsetEval` evaluator coupled with the BestFirst search method.

- Evaluation mode

  We evaluated the selection on the full training set. At this stage, our goal was exploratory feature discovery across the entire sample rather than evaluating model generalizability (which is reserved for the algorithm testing phase via 10-fold cross-validation).

- Findings

  Out of 24 encoded features, the evaluator identified a subset of just 5 attributes that produced a highly significant predictive merit of 0.937 (93.7%). The process confirmed that the following attributes have the highest correlation with the target variable (price):

  1. class (top global predictor)
  2. duration
  3. days_left
  4. airline=Vistara
  5. source_city=Delhi

3. Data Splitting

Instead of manually creating separate training and testing files, we used Weka's built-in evaluation procedures within the Classify module. This approach ensures that the data is partitioned dynamically and randomly, reducing the risk of selection bias.

3.1 10-fold Cross-Validation

This was our primary method for evaluating model performance. By dividing the dataset into ten folds and rotating the test set, we ensured that the final error metrics (such as Correlation Coefficient and MAE) are robust and representative of the entire data distribution.

3.2 Percentage Split (80%)

We also used the percentage split option to verify the stability of the models. In this mode, Weka automatically allocated 12,005 instances for training and held out 3,001 for final testing.

4. Algorithm Selection

Based on the nature of our target variable (price), which is a continuous numeric value, the core task of this project is **Regression** rather than Classification. To ensure a comprehensive analysis, we selected three distinct algorithms representing different learning paradigms to compare their predictive performance.

4.1 Linear Regression

We selected Linear Regression as our primary model. Its main advantage is high interpretability, allowing us to clearly see the weight (coefficient) assigned to each feature. It assumes a linear relationship between the independent variables (e.g., days_left, duration) and the target variable. If the underlying data structure is simple, this algorithm provides a fast and effective solution.

4.2 Random Forest (Decision Trees)

To handle potential non-linear relationships and complex interactions between features, we selected Random Forest. Unlike a single decision tree, this method builds multiple trees during training and outputs the average prediction. It is highly robust to remaining noise in the dataset and excels at finding sudden price jumps (e.g., when prices spike because a booking deadline is reached or cheap seats sell out), rather than forcing the data into a strictly straight line.

4.3 SMOreg (Support Vector Machines for Regression)

Finally, we chose SMOreg, Weka's implementation of Support Vector Regression (SVR). This algorithm maps the data into a higher-dimensional space to find the optimal regression hyperplane. It is particularly effective on our dataset because we applied the `Normalize` filter during preprocessing; SMOreg relies heavily on distance calculations and performs best when all numerical features are scaled equally.

5. Train and Test Model: Results and Evaluation

To rigorously assess the predictive capabilities of our selected algorithms, we evaluated them using our preprocessed subset of 15,006 instances and the most 5 significant attributes identified during feature selection. We used both 10-fold Cross-Validation (our primary metric) and an 80% Percentage Split to ensure the results were not biased by a specific training set.

The evaluation metrics focused on the Correlation Coefficient (closer to 1 indicates a perfect linear relationship between predicted and actual prices) and the Mean Absolute Error (MAE) (lower values indicate smaller average prediction errors).

5.1 Performance Overview (10-fold Cross-Validation)

Algorithm	Correlation Coefficient	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)
Linear Regression	0.9459	0.0399	0.0652
SMOreg (SVM)	0.9447	0.0385	0.0660
Random Forest	0.9693	0.0292	0.0494

Note: The 80% Percentage Split tests produced near-identical metrics across all models, confirming that the models are highly stable and not suffering from significant overfitting.

5.2 Model Analysis

1. *Linear Regression - the efficient baseline*

   This algorithm performed admirably, achieving a correlation coefficient of 0.94. Its greatest strength was extreme computational efficiency, training in just fractions of a second. Looking at the Linear Regression equation generated by Weka, we can definitely confirm our initial hypothesis regarding feature weights. The model clearly shows that **class** has the strongest positive impact on price (+0.3922 weight), while days_left has a strong negative correlation (-0.0507 weight, meaning price decreases as days_left increase).

2. *SMOreg - the computational bottleneck*

   While SMOreg achieved accuracy metrics virtually identical to Linear Regression (Correlation 0.944), it suffered from a severe computational processing time. Training the model with 10-fold cross-validation took approximately **40 minutes**. This exponential increase in build time occurs because Support Vector Machines (like SMOreg) must calculate the distance between every single instance pair in a higher-dimensional space to define the optimal regression hyperplane. Given that it provided no accuracy improvement over the Linear Regression, SMOreg proved **inefficient** for this specific dataset size.

3. *Random Forest - the champion*

   The Random Forest model significantly outperformed the others. With a Correlation Coefficient of nearly 0.97 and the lowest error rates (MAE = 0.0292), it proved its superiority. The main reason for this success is its ability to find non-linear patterns (as hypothesized in section 4). Flight prices do not increase gradually day by day. Instead, they experience sudden jumps whenever cheaper seats sell out or a booking deadline is reached. The Random Forest model is excellent at capturing these sudden "steps" mathematically. Furthermore, even though it builds a complex set of multiple trees, it completed the 10-fold CV training in less than a minute, proving to be the **best choice** for balancing high accuracy with a computational speed.

6. Splitting into Business and Economy

Our initial model results from Chapter 5 showed that the class attribute completely dominated, masking the subtle predictive interactions of secondary features. To get further insights into the distinct commercial dynamics governing flight pricing in India, the dataset was segmented into two separate sub-datasets: Economy Class (10,329 instances) and Business Class (4,676 instances).

The number of instances in the Business class subset is notably smaller than that of the Economy class subset. This asymmetric reduction is intentional; by bypassing standalone sampling for each tier, we ensure that the segmented algorithms are trained and tested on the exact same data we used for our previous model..

To ensure methodological consistency and permit a direct comparison of the predictive outcomes, the exact preprocessing pipeline established in Chapter 2 was replicated across each isolated data split. Following the segmentation of the dataset, the class attribute was physically removed from the data pools using the `Remove` filter.

6.1 Economy Class Models and Preprocessing Variations

The isolated Economy class subset was evaluated across two distinct methodologies: first, by maintaining the raw normalized scale of the target variable, and second, by executing a non-linear logarithmic transformation. During the preparation of this sub-dataset, the Interquartile Range (IQR) filter identified and physically pruned two outliers using the `RemoveWithValues` pipeline.

6.1.1 Baseline Economy Models (Without Log Transformation)

Rerunning Correlation-based Feature Selection directly over the raw-price Economy segment yielded a baseline subset predictive merit of 0.613. The evaluator identified an optimal subset of five attributes: airline=AirAsia, airline=Vistara, source_city=Kolkata, stops, and days_left.

The mathematical expression generated by Weka for the baseline Linear Regression model is as follows:

*price = -0.0643 * airline=AirAsia + 0.0403 * airline=Vistara + 0.0288 * source_city=Kolkata + 0.1463 * stops - 0.2155 * days_left + 0.1989*

This un-logged configuration returned moderate performance metrics under 10-fold cross-validation:

- <u>Linear Regression</u> achieved a correlation coefficient of 0.6835 with a Mean Absolute Error (MAE) of 0.0590
- <u>SMOreg</u> achieved a comparable cross-validation correlation coefficient of 0.6802 and an MAE of 0.0574. However, it introduced a severe computational processing penalty, requiring **292.48 seconds** to complete training due to evaluating 803,098,918 kernels across the raw linear target space.
- <u>Random Forest</u> easily beat out both of the other models, giving us the highest accuracy out of the baseline models with a correlation coefficient of 0.7788 and an **MAE of 0.0468**.

6.1.2 Optimized Economy Models (With Log Transformation)

We noticed a heavily **right-skewed pricing profile for the Economy subset**. The vast majority of standard tickets cluster tightly around affordable base rates, whereas immediate, last-minute booking frames create exponential, non-linear fare spikes. To stabilize the target variance, the unsupervised `MathExpression` filter was used to execute a logarithmic transformation on the dependent variable:

E = log(A)

Doing attribute selection across this log-transformed target space changed which attributes were selected. The global merit of the best subset advanced from **0.613 to 0.669**, and a sixth predictor - **duration** - became important.

The mathematical expression computed by the log-scaled Linear Regression model is formulated as follows:

*price = -0.1266 * airline=AirAsia + 0.0552 * airline=Vistara + 0.0489 * source_city=Kolkata + 0.1966 * stops + 0.1110 * duration - 0.2775 * days_left + 0.4892*

The application of the logarithmic transformation universally improved the accuracy across the board:

- Linear Regression**:** the 10-fold cross-validation correlation coefficient surged to 0.7387 (a clear improvement of +0.0552), while the Relative Absolute Error (**RAE**) dropped from 72.40% down to **68.02%**.
- SMOreg (SVM): the cross-validation correlation leaped to 0.7377; crucially, flattening the right-skewed target distribution reduced time to compile the model to just **63.24 seconds** - saving nearly four minutes compared to the un-logged execution.
- Random Forest: remained the absolute performance leader for this segment, pushing its 10-fold cross-validation correlation coefficient to 0.8074 (and climbing to **0.8141** on the 80% test split) while minimizing its relative error metrics.

6.2 Business Class Models

The Business class sub-dataset was also processed using the exact baseline pipeline features established in Chapter 2. The IQR filter returned **zero outliers** within this partition, signaling a more stable premium distribution profile. It is also of importance to note that only two airlines offer business class tickets: Air_India and Vistara.

A logarithmic target transformation was intentionally not used for the Business class. Unlike the low-cost and dynamic prices of Economy tickets, Business class airfares offer a symmetric distribution that is concentrated within a certain price range.

The Correlation-based Attribute Selection evaluator returned a baseline subset merit of 0.596, identifying six optimal attributes: airline=Vistara, source_city=Kolkata, stops, arrival_time, destination_city=Delhi, days_left.

The resulting Business Class Linear Regression mathematical expression is detailed below:

*price = 0.0744 * airline=Vistara + 0.0330 * source_city=Kolkata + 0.4791 * stops + 0.0280 * arrival_time - 0.0216 * destination_city=Delhi - 0.0351 * days_left + 0.1243*

Evaluating the Business class models revealed a prominent convergence in accuracy metrics across all three machine learning frameworks:

- Linear Regression: achieved a steady cross-validation correlation coefficient of 0.6726 and an MAE of 0.0726, which advanced to 0.6970 correlation on the 80% percentage split validation.
- SMOreg (SVM): mirroring the linear baseline, SMOreg tracked a 10-fold cross-validation correlation coefficient of 0.6695 and an MAE of 0.0724, computing efficiently in **17.85 seconds** due to the reduced instance size.
- Random Forest: unlike its behavior in the Economy and combined models, Random Forest experienced an accuracy plateau in the Business segment; it essentially matched the linear frameworks, producing a cross-validation correlation coefficient of 0.6732 and an MAE of 0.0731.

7. Insights and Limitations

7.1 Base Model (Both classes)

Our first round of models trained on the combined dataset returned incredibly high accuracy , with the Random Forest model achieving a correlation coefficient of 0.9693 and a low Mean Absolute Error (MAE) of 0.0292. However, an analysis of the feature weights revealed that this predictive confidence was almost entirely driven by the class attribute.

Because a Business Class ticket costs a great deal more than an Economy ticket, the global models primarily functioned as class discriminators rather than mapping subtle pricing variations. Any variance introduced by booking windows or flight duration was statistically overshadowed by this massive premium price gap.

Furthermore, the global dataset exposed a major computational constraint for Support Vector Regression: Weka's SMOreg required approximately **40 minutes** to complete a 10-fold cross-validation. This cross-validation timeline proved highly inefficient, especially since it offered no accuracy improvement over the basic Linear Regression baseline (**0.9459** correlation), which compiled in fractions of a second.

7.2 Class Isolation

7.2.1 Economy Class

Analysing the Economy Class separately immediately showed how the global models hugely rely on the class attribute. Upon removing it, the feature selection merit dropped significantly from **0.937 to 0.613**, proving that the models had to work a lot harder to explain price variance without the binary class divider.

Comparing the raw models to the log-transformed ones highlighted a few clear trends:

- **Urgency:** In the raw baseline Economy model, booking proximity held a strong negative coefficient of -0.2155; once we stabilized the target distribution via logarithmic transformation, this weight sharpened to **-0.2775**; this mathematical shift shows that booking urgency dominates the economy market; as the departure window closes, fares increase at an exponential rate.
- **Hidden influence:** Under the raw linear scale, the impact of flight length was not recognized as an important attribute; transforming the target scale logarithmically allowed the attribute selector to identify **duration** as a significant predictive attribute, increasing the overall subset merit to 0.669.
- **Algorithmic optimizations:** The logarithmic transformation flattened the volatile target variance, yielding an absolute correlation increase of +0.0552 for Linear Regression; it also improved the SMOreg's build latency from **292.48 seconds to 63.24 seconds**.

7.2.2 Business Class

The business class segment works in completely the opposite way compared to economy. The resulting models revealed a convergence in accuracy, with Random Forest (0.6732 correlation) flattening completely and matching the basic Linear Regression baseline (0.6726).

The biggest insight was the difference in feature importance:

- **Routing Structure:** In stark contrast to the Economy segment, the Business Class model assigned **stops** an overwhelming positive coefficient of **+0.4791**. On the other hand, booking proximity dropped to a negligible negative weight of **-0.0351** for days_left, meaning that the urgency does not play a big role in the pricing. This is probably caused by a higher availability of business tickets even close to the flight date.
- **The Business Traveler Profile:** The trained models reflect the real-world priorities of premium passengers. Corporate travelers frequently book flights at a moment's notice based on immediate business needs, allowing airlines to maintain high premium base rates regardless of the remaining booking window. Business travelers prioritize time over fare flexibility, showing an extreme willingness to pay a high price for direct travel. Because of this, airlines can heavily mark up multi-stop corporate flights to maximize their profits.
- **Airline and Service Profile Shifts:** This corporate alignment is further verified by the structural variables selected by Weka. Carriers without competitive business cabins on these routes dropped entirely out of the feature matrix. Instead, they are replaced by features that matter to corporate travelers, such as chronological **arrival_time** (+0.0280) and high-profile business destinations like **destination_city=Delhi** (-0.0216), proving that convenience and location drive premium ticket costs.

7.3 Project Data and Structural Limitations

Despite achieving robust predictive capabilities across our segmented models, several data boundaries restrict the scope and universal generalizability of these findings:

- **Temporal Constraints:** The data is a narrow 50-day snapshot collected via web scraping. Because of this short window, our models cannot account for broader seasonal shifts, summer/winter vacation drop-offs, or major Indian holiday periods like Diwali or Holi. During these times, travel demand spikes heavily,

and airline pricing algorithms behave completely differently than during standard weeks.
- **Geographic Constraints:** The analysis is bound strictly within flights connecting six primary metropolitan transit hubs (Delhi, Mumbai, Bangalore, Kolkata, Hyderabad, and Chennai). By limiting our analysis to these massive airports, we miss out on regional tier-2 and tier-3 routes, which operate under lower infrastructure capacity, completely different competitive environments, and unique regional supply-demand dynamics.
- **Market Concentration:** In the premium business class sector, the data is heavily dominated by just two legacy airlines: Air India and Vistara. This lack of market variety means our business class equations are inevitably biased toward the specific corporate pricing models and strategies of these two companies, rather than a diverse, highly competitive premium market.
- **Binary Class Simplification:** The dataset reduces cabin choices to a simple binary variable (Economy vs. Business). This leaves out real-world intermediate options like Premium Economy or top-tier First Class, which introduce additional price variations that our models were unable to capture.
- **The Temporal Paradox:** A surprising finding during feature engineering was that our binned time metrics (departure_time and arrival_time) showed far less statistical importance than structural attributes like physical flight duration or stops. This creates a bit of a paradox, as departure and arrival times are usually major priorities for real-world corporate travelers who need to stick to tight schedules.

7.4 Conclusions

This project used several regression models to break down and understand what actually drives domestic airfare prices in India. Our main finding is that analyzing an entire aviation dataset as one giant block introduces a huge amount of structural bias. While our global combined model looked highly accurate on paper, that accuracy was mostly a shortcut - it was just exploiting the massive, five-fold price gap between economy and business class tickets. By merging them together, the model completely missed the unique variables that matter to each individual market.

By splitting the dataset by cabin class, we proved that economy and business sectors operate under two completely different commercial logics. The economy sector is heavily driven by booking urgency; prices don't rise steadily, but rather spike exponentially as a multiplicative function of how close you get to the departure date. Business class, on the other hand, follows a strict linear structure where booking deadlines barely matter. Instead, corporate routes are priced based on flight efficiency - meaning airlines charge a heavy premium to completely avoid intermediate layovers - and how well the flight time fits a standard business day.

Ultimately, our choice of algorithms showed that while complex tree-based models like Random Forest are necessary to capture the volatile price jumps in the economy market, traditional linear models are perfectly fine and much faster for predicting the highly uniform, linear trends found in business class travel.

7.5 Recommendations and Suggestions for Improvement

Looking back at the data limitations and how our different models behaved, we have a few clear recommendations for how a future project could expand on this work:

- **Longitudinal Data Expansion:** Instead of relying on a short, isolated snapshot, future projects should try to collect multi-year, longitudinal flight data. Tracking how ticket prices move across a full 360-day calendar year would let the regression models clearly separate short-term holiday demand spikes from the normal, baseline price adjustments.
- **Regional Route Modeling:** We recommend expanding the geographic scope of the data to include regional flights connecting smaller tier-2 and tier-3 cities. Modeling the pricing structures of these low-capacity regional airports would give a much more well-rounded and generalizable view of domestic travel economics, rather than just looking at the busiest metropolitan hubs.